

YUBO ZHAO

Software Engineering | Machine Learning | Quantitative Research | Robotics

Waterloo, Ontario | bowenzhao2020@gmail.com | +1 418 919 8188

github.com/YuboZhao2008 | linkedin.com/in/yubozhao-ai | Portfolio

EDUCATION

University of Waterloo - Bachelor of Software Engineering

Expected 2031

1A Software Engineering | Waterloo, Ontario

EXPERIENCE & COMPETITIONS

Team Canada Member - International Olympiad in Artificial Intelligence

2025

- Selected to represent Canada at IOAI 2025; prepared in deep learning, computer vision, NLP, data preparation, and model optimization for competitive AI problems.

Robotics Team Member - FIRST Tech Challenge World Championship

2024

FTC World Championship - 2nd Place Globally (team result)

- Collaborated on autonomous and driver-controlled robot development, integrating sensors and control logic through iterative debugging and performance tuning.

Machine Learning / Quantitative Research Intern

Jul-Aug 2025

Shaanxi Shun Yicheng Investment Management Co., Ltd. | Xi'an, China

- Supported financial ML workflows spanning preprocessing, feature engineering, model evaluation, signal validation, and benchmarking; investigated backtesting, leakage prevention, and overfitting control.

SELECTED PROJECTS

Warcraft III Reinforcement Learning Agent | Python | Lua | RL | PPO

- Built bidirectional Python-Lua IPC for Warcraft III: Reforged, with a sequential action queue to synchronize external agent commands and work around stale command/file caching.
- Instrumented live HP, position, distance, targets, and combat state in Lua; exposed discrete movement, stop, targeting, attack, and hero-ability actions to Python.
- Designed Gymnasium-style observation/action/reward spaces and a PPO-oriented architecture for 1v1 hero combat, with rewards for damage dealt/received, kills/deaths, survival, and positioning.

Brain Tumor Detection System | Python | PyTorch | MiniMaxViT

- Developed an MRI classification pipeline using a compact vision transformer; tested augmentation, oversampling, class balancing, and validation strategies.
- Evaluated precision, recall, F1, confusion matrices, generalization, and potential dataset artifacts.

Gesture-Controlled Drawing Interface | Python | OpenCV | MediaPipe | scikit-learn

- Trained a Random Forest on 21-point hand landmarks to classify open, closed, and pointing gestures with 98% accuracy on the project test split.
- Implemented real-time drawing and erasing with temporal smoothing, gesture-state logic, latency controls, tool switching, and snapshot capture.

Jarvis - Local Multimodal AI Assistant | Python | Ollama | OpenCV | APIs

- Built a modular local assistant integrating LLM inference, speech-to-text, text-to-speech, memory, external APIs, and real-time vision across voice, text, and camera input.

TECHNICAL SKILLS

Languages & Software: Python, Lua, C++, Java, JavaScript, SQL, HTML/CSS, Git, Linux, REST APIs

ML & Data: PyTorch, scikit-learn, NumPy, pandas, Matplotlib; vision transformers, transfer learning, feature engineering, cross-validation, reinforcement learning environment design

AI & Robotics: OpenCV, MediaPipe, Ollama, multimodal systems, speech recognition, sensors, PID control

HONOURS

Euclid 2026 - Honour Roll, 86/100 | CSMC 2025 - Honour Roll, 48/60

Sir Isaac Newton Physics Contest - Top 5% Globally